Technical Appendix

Employment Protection and the Disability Employment Gap

This appendix documents the estimators, assumptions, executed comparisons, and verification procedures behind the accompanying paper. The acquired source is the repository author's GitHub release containing SAS data and a checked July 2001 paper proof; it contains no original SAS analysis programs. Publication descriptions, source labels, and numerical checks support a reconstruction, with unresolved differences retained. This is not a claim of source-code equivalence or a verified complete acquisition of the linked Dataverse deposit. The potential-outcomes framework follows Ding (2024). Software tests and numerical agreement establish different kinds of computational evidence; neither validates the causal identifying assumptions.

1 A common target for four repeated-cross-section cells

Let $G \in \{0,1\}$ denote reported work-limiting disability, $T \in \{0,1\}$ the pre/post period, X the selected covariates, and Y an annual employment outcome. Write $A = (G,T)$ and abbreviate its four values by the **two-character cell labels** $00, 01, 10, 11$, rather than the product $G \times T$. The target cell is $a^* = 11$. Different people are observed across years, so the analysis never constructs an individual outcome change from unmatched records.

Let $s_i > 0$ be the analysis weight. Within each sex-by-age analysis sample, the pipeline divides the original survey weight by its year's total and then by the number of years in that period. Thus years have equal mass within a period, and the pre and post periods each have total mass one. Survey weights define the within-year population; this additional normalization declares the mixture of years. It does not create a panel or a survey-design variance estimator. The annual descriptive and event-study analyses retain the original relative survey weights instead.

Use E_s and P_s for expectations and probabilities in this declared weighted population. Define

$$q_a(x) = P_s(A = a \mid X = x), \qquad m_a(x) = E_s[Y \mid A = a, X = x], \qquad p_* = P_s(A = 11).$$

For every source cell, the common-target mean is

$$\mu_a = E_s[m_a(X) \mid A = 11].$$

The target mean $\mu_{11} = E_s[Y \mid A = 11]$ is observed. The statistical estimand is

$$\theta = \mu_{11} - \mu_{10} - \mu_{01} + \mu_{00}.$$

All adjusted methods use the same complete-case observations, period windows, covariates, and disabled-post target. The unadjusted DID is also reported, but its four raw means average over each cell's own covariate distribution; it is a descriptive benchmark rather than another estimator of exactly the same standardized functional when composition differs.

1.1 Transport, standardization, and augmentation

Let $S_* = \sum_i s_i 1(A_i = 11)$ and $\widehat{r}_a(x) = \widehat{q}_{11}(x)/\widehat{q}_a(x)$. For a source cell $a \in \{00, 01, 10\}$, regression standardization is

$$\widehat{\mu}_a^{\mathrm{RA}} = \frac{1}{S_*} \sum_i s_i 1(A_i = 11) \widehat{m}_a(X_i).$$

The normalized IPW transported mean is

$$\widehat{\mu}_a^{\mathrm{IPW}} = \frac{\sum_i s_i 1(A_i = a) \widehat{r}_a(X_i) Y_i}{\sum_i s_i 1(A_i = a) \widehat{r}_a(X_i)}.$$

The implemented augmented mean instead uses target-mass normalization in its residual correction:

$$\widehat{\mu}_a^{\mathrm{AIPW}} = \widehat{\mu}_a^{\mathrm{RA}} + \frac{1}{S_*} \sum_i s_i 1(A_i = a) \widehat{r}_a(X_i) \{Y_i - \widehat{m}_a(X_i)\}.$$

For $a = 11$, use the weighted observed mean for all methods. Insert the four means into the same signed contrast to obtain each adjusted DID. The IPW and AIPW denominators deliberately differ: replacing S_* by the transported source mass in the residual term produces a different finite-sample estimator. Multiplying every s_i by a common positive constant leaves all these estimates unchanged.

The initial nuisance models are a weighted multinomial logit for the four cell probabilities and separate weighted linear outcome models in the three source cells. The primary covariates are centered age, its square, and the historical recorded race categories, with analyses stratified by recorded sex and age range. The richer specification additionally adjusts for recorded education and region. These additional variables are not assumed automatically to be unaffected by policy. Neither model adjusts for benefits receipt, current employment, or another realized labor-market outcome as a baseline confounder.

This construction is a common-target transport contrast. It is not presented as an implementation of the particular repeated-cross-section score in Sant'Anna and Zhao (2020). That literature and the semiparametric DID framework of Abadie (2005) motivate adjustment under explicit identifying assumptions; method names alone do not establish those assumptions.

1.2 Why the augmentation is doubly robust

Suppose the fitted source outcome model and density ratio converge to $\widetilde{m}_a$ and $\widetilde{r}_a$, respectively. The population limit of the augmented source mean is

$$\widetilde{\mu}_a = \frac{E_s[1(A = 11)\widetilde{m}_a(X)]}{p_*} + \frac{E_s[1(A = a)\widetilde{r}_a(X)\{Y - \widetilde{m}_a(X)\}]}{p_*}.$$

Iterated expectation gives the bias identity

$$\widetilde{\mu}_a - \mu_a = \frac{1}{p_*} E_s[\{q_{11}(X) - q_a(X)\widetilde{r}_a(X)\}\{\widetilde{m}_a(X) - m_a(X)\}].$$

The bias vanishes if the source outcome model is correct or the density ratio is correct. For the full contrast, the condition must hold for **each** of the three source cells. Merely fitting both models does not guarantee that either is correct. Consistency also requires support, integrability, and suitable convergence conditions. Flexible models would require additional attention to overfitting, cross-fitting, and rates; the current parametric implementation does not claim those features.

Double robustness concerns estimation of the defined observed-data functional. It does not make conditional parallel trends true, stabilize changing disability reporting, or turn a post-policy characteristic into a baseline covariate.

2 Conditions for a policy interpretation

Let G^* describe a stable, pre-policy disability group and let $Y_t(p)$ denote the outcome under policy regime $p \in \{0, 1\}$. An ATT-like interpretation of the standardized contrast requires a link between observed G and this stable group, as well as the following conditions.

1. **Consistency and a defined exposure.** Recorded outcomes correspond to the policy regime received, with an account of phased employer coverage and partial exposure. The ADA is the intervention; disability itself is not.
2. **No anticipation in the chosen pre window.** The pre outcomes represent the no-policy regime. Passage, implementation, and a first full outcome year must not be treated as interchangeable dates.
3. **Conditional parallel untreated changes.** For relevant x,

$$E[Y_1(0) - Y_0(0) \mid G^* = 1, X = x] = E[Y_1(0) - Y_0(0) \mid G^* = 0, X = x].$$

   Repeated-cross-section sampling must permit identification of the two period-specific conditional means for each stable group. The comparison also needs a defensible restriction on spillovers or other policy effects on the comparison group.
4. **Stable composition and measurement.** Changes in disability reporting, sample inclusion, measurement, or unobserved health must not independently change the group contrast after adjustment. Conditional standardization cannot correct unmeasured changes in who is counted as disabled.
5. **Common support and appropriate covariates.** Every target covariate pattern must be supported in all source cells. Covariates used to justify identification must be predetermined or otherwise justified, rather than chosen only because they improve prediction.

Under these restrictions, the change in the comparison group's untreated outcomes supplies the missing change for the stable treated group. Without them, θ remains a standardized change in observed group disparities. Survey precision and agreement among RA, IPW, and AIPW cannot distinguish these interpretations by themselves. Pre-period event-study diagnostics are useful but have limited power, and selecting an analysis after a pre-trend test can distort inference (Roth 2022).

2.1 Reporting endogeneity as a causal graph

Consider predetermined characteristics X_0, underlying health H_0, policy P, annual employment Y_t, disability reported at a later interview G_{t+1}, and contemporaneous benefits receipt B_{t+1}. A plausible graph includes the following directed paths:

$$\begin{aligned} &P \longrightarrow Y_t \longrightarrow G_{t+1}, && P \longrightarrow G_{t+1}, \\ &H_0 \longrightarrow Y_t, && H_0 \longrightarrow G_{t+1}, \\ &X_0 \longrightarrow H_0, && X_0 \longrightarrow Y_t, \\ &P \longrightarrow B_{t+1}, && Y_t \longrightarrow B_{t+1} \longleftarrow H_0. \end{aligned}$$

These edges describe threats to assess, not estimated causal relationships. Indexing employment before interview reporting avoids a simultaneous feedback cycle. The observed disability group can be a descendant of policy and employment. Conditioning on it can also select on the collider $P \to G_{t+1} \leftarrow H_0$. Benefits receipt is likewise an unsafe automatic control or sample restriction in

this graph. Excluding recipients can select on employment and health rather than isolate a pure policy effect.

An observed decline in disabled-group employment could therefore reflect changes in reporting or composition even if the policy's effect on any fixed person were zero. The contribution of this study is to make that possibility visible through timing, prevalence, composition, and sensitivity checks, not to claim that reweighting eliminates it.

3 Diagnostics, uncertainty, and resampling

3.1 Support and balance

The initial software draft used 0.01 as a lower bound on fitted four-cell joint probabilities. Inspection of the acquired data showed that this guard conflated rare disability-cell prevalence with the covariate support required for transport. The amended empirical runner uses 10^{-6} as a numerical floor and retains an effective-sample-size minimum of 20. It reports the target-weight share below 0.01 rather than silently removing those observations. The same amended rule applies to every empirical stratum, window, outcome, and bootstrap refit; the generic function's default remains 0.01. The chronology, rationale, and diagnostic evidence are recorded in `docs/support-amendment.md`.

For transported weights v_i, the reported quantity is $\text{ESS} = (\sum_i v_i)^2 / \sum_i v_i^2$; it is a weight concentration diagnostic, not the number of independent survey clusters. The maximum normalized weight share and cell sample counts accompany it. These thresholds are declared computational diagnostics, not universal criteria for causal validity. Failure stops that specification; the software does not silently clip probabilities or remove observations.

To separate prevalence from relative covariate support, it also reports

$$\rho_a(x) = \frac{q_{11}(x)}{q_a(x)} \frac{P_s(A = a)}{P_s(A = 11)} = \frac{f_s(x \mid A = 11)}{f_s(x \mid A = a)}.$$

The fitted joint probabilities, density ratios, ESS, and maximum weight shares must be assessed together. None tests conditional parallel trends or stable disability reporting.

Balance compares each donor cell with the same disabled-post target before and after transport. Standardized differences use the target population's weighted standard deviation for each model-matrix column. A target-constant column has an undefined standardized difference and is flagged rather than treated as perfectly balanced. Balance on included columns cannot demonstrate balance on unobserved health or validate the counterfactual trend.

3.2 Shared covariance and the meaning of uncertainty

The four estimates share observations through the target distribution, multinomial model, and nuisance fitting. Their standard errors cannot be added as though four independent cell means were being combined. For illustration, with fixed correct nuisances and independent observations, let $d_* = E[s1(A = 11)]$. A source-mean influence contribution takes the form

$$\phi_a = \frac{s}{d_*} \left[1(A = 11)\{m_a(X) - \mu_a\} + 1(A = a) r_a(X)\{Y - m_a(X)\} \right],$$

while $\phi_{11} = s1(A = 11)(Y - \mu_{11})/d_*$. The contrast contribution is $\phi_\theta = \phi_{11} - \phi_{10} - \phi_{01} + \phi_{00}$, whose variance includes all covariances. Estimated nuisance models and a complex sample design require additional justification; this display is not the implemented variance formula.

The annual event-study intervals use HC1 under an individual-independence working model. They do not account automatically for CPS rotation, household dependence, the sample design, or a common national disability-specific shock. Serial dependence can seriously distort DID uncertainty (Bertrand et al. 2004). There is one national policy timeline, so a large number of survey respondents does not create many independent policy assignments. Neither a state-cluster standard error nor the individual bootstrap below can remove a confounding nationwide trend.

3.3 Implemented bootstrap algorithm

For a fixed outcome, stratum, window, covariate specification, and complete-case sample, the optional bootstrap proceeds as follows:

1. Fit the point estimator and retain the original analysis weights and the observed cell counts $n_{00}, n_{01}, n_{10}, n_{11}$.
2. Within each cell independently, draw n_a rows with replacement. Carry each selected row's analysis weight with it. This conditions on the observed four cell counts; it does not resample a CPS survey design.
3. Refit the multinomial model and all three source outcome regressions on the resampled observations. Recompute the observed target mean, adjustment estimates, and support diagnostics. Resampling only precomputed treatment effects or holding the fitted nuisances fixed would omit uncertainty.
4. Repeat using the recorded seed. Save every replicate's status and any failure reason. By default, **all requested replicates must succeed** before bootstrap standard errors or intervals are reported. An explicitly relaxed success requirement is labeled as inference conditional on successful replicates, not silently substituted for the default.
5. For an eligible run, report the replicate standard deviation and the 2.5th and 97.5th percentiles using R's type-7 quantile convention. Runs with fewer than 199 replicates are labeled exploratory. Even 199 replicates gives limited tail resolution; more draws address Monte Carlo error, not a wrong sampling or identification model.

Reproducible parallel execution assigns a seed to each replicate before dispatch. The serial and parallel implementations therefore produce the same draws and results for a fixed seed. Multinomial fitting combines identical covariate/cell rows by summing their weights, an exact likelihood reduction; no respondent subsample replaces the declared empirical population.

The main pipeline requests this procedure for the primary-window weeks-worked and any-work analyses in each of the four sex-by-age strata. The bootstrap conditions on the supplied period/year analysis weights rather than rebuilding their original survey totals in every draw. These intervals are therefore conditional sampling summaries. A final design-based analysis requires the actual archive's household/PSU/stratum information or appropriate replicate weights and a resampling method consistent with them.

4 Distributional and trend sensitivity extensions

4.1 Threshold outcomes are not quantile treatment effects

For $k \in \{0, 13, 26, 39, 51\}$, define $Y_k = 1(\text{weeks} \leq k)$. The same four-cell method estimates

$$\theta_k = F^*_{11}(k) - F^*_{10}(k) - F^*_{01}(k) + F^*_{00}(k),$$

where $F_a^*(k)$ is the source cell's cumulative employment distribution standardized to the disabled-post target. At $k = 0$, the outcome is no weeks worked; higher thresholds examine changes in the share with relatively little annual employment. A positive θ_k is an increase in the cumulative probability contrast, not an increase in weeks worked. Since the samples are the same and the estimator is linear in the outcome, the zero-weeks contrast is the negative of the any-work contrast, up to numerical precision.

A causal distributional interpretation requires parallel trends for the threshold indicators, which does not follow from parallel trends in mean weeks. These contrasts neither identify individual rank preservation nor quantile treatment effects. The implied counterfactual CDF need not satisfy range and monotonicity restrictions under arbitrary data and models; it should not be inverted mechanically. Thresholds are prespecified, and a collection of pointwise comparisons must not be described as a simultaneous confidence band. The current distributional output is a diagnostic point-estimate extension, not a quantified causal distributional claim.

4.2 Exact linear loading for the implemented bias curve

Let $\mathcal{L}(Y)$ denote the fitted AIPW contrast for fixed observations, weights, covariates, and cell-probability estimates. Because the source outcome fits are weighted linear regressions with fixed design matrices, this estimator is linear in Y. Define

$$H_i = G_i(\text{outcome_year}_i - 1987), \qquad L = \mathcal{L}(H), \qquad \widehat{\theta} = \mathcal{L}(Y).$$

For a hypothetical differential untreated drift of b outcome units per year, the outcome perturbation $Y_i^{(b)} = Y_i - bH_i$ gives the exact fitted-model identity

$$\mathcal{L}(Y^{(b)}) = \widehat{\theta} - bL.$$

The pipeline computes L by running the **same estimator** on H. It is not replaced by the arithmetic difference between mean pre and post years: common target standardization and estimated residual corrections can change that loading. The output column named `year_separation` stores this estimator loading. With $L \neq 0$, the drift that moves the point contrast to zero is $b_0 = \widehat{\theta}/L$; it is undefined when the loading vanishes.

The declared grid is $b = -1, -0.75, \ldots, 1$ weeks per year for weeks worked and $b = -0.02, -0.015, \ldots, 0.02$ probability units per year for any work. Applying this algebra to a binary outcome is a bias sensitivity calculation, not a claim that every perturbed value is a realizable binary observation. Holding the sample and nuisance designs fixed is essential to the identity. Tests compare the shortcut against direct outcome-perturbation refits.

This is a one-dimensional, linear trend-bias model. It is not a general identification region, a simultaneous sensitivity interval, or an implementation of the robust trend restrictions in Rambachan and Roth (2023). Its role is to put the counterfactual trend required to explain the point estimate on an interpretable scale. Sampling uncertainty and uncertainty about b remain distinct.

5 Executed reconstruction and additional results

5.1 Scope of the numerical comparison

The supplied SAS extract contains 971,058 records. Age eligibility retains 759,771; excluding missing or zero `FNLWGT2` leaves 709,125 weighted estimation records. Table 1's printed observation

counts refer to the raw age-eligible population, whereas its weighted means and the regressions use valid weights. Keeping these populations separate reconciles all 40 selected Table 1 counts and all eight Table 2 estimation counts. Original weight-construction programs are absent, so the precise historical reason for each excluded weight is not asserted.

The reconstructed Table 2 no-trend model uses outcome-year 1987 as the reference, full categorical demographic controls, and year-by-demographic interactions. This differs from the extension's age-polynomial event-study model. Its conventional WLS covariance is reported as the reconstruction convention; the extension's HC1 and bootstrap uncertainty have different interpretations. The benchmark source is the supplied July 2001 proof, not an independently retrieved final printing. Proof/PDF pages 13–14 contain Table 1, and page 17 contains Table 2.

The executed comparison matches **265 of 272 selected entries** at their declared printed-precision tolerances. **7 discrepancies remain**; they are retained below rather than relabeled as passes. The selection covers Table 1 age/weekly-employment means and counts, and Table 2 no-trend employment/log-wage coefficients, standard errors, and counts. It does not cover every original table or the trend columns.

Table 1: All retained discrepancies. T1 uses survey years; T2 uses outcome years. M/F denote recorded male/female strata.

Comparison	Year	Printed	Reconstructed	Tolerance
T1 F40–58 weeks mean	1996	11.300	11.755862	5e-02
T2 M40–58 weeks SE	1988	0.550	0.586078	5e-03
T2 M21–39 log wage SE	1991	0.043	0.043529	5e-04
T2 F21–39 log wage coef.	1992	-0.010	-0.009175	5e-04
T2 F21–39 log wage SE	1993	0.050	0.051201	5e-04
T2 M40–58 log wage SE	1993	0.042	0.043269	5e-04
T2 F40–58 log wage SE	1994	0.051	0.048878	5e-04

`docs/replication-audit.md` records the complete comparison rules, source transcription, and independent checks. All 36 selected Table 2 employment coefficients match the source precision. The disputed older-male 1988 weeks standard error and young-female 1992 wage coefficient were also reproduced by independent `stats::lm` fits on the actual samples. These checks make a package-specific numerical error less likely, but do not establish that the proof's values are typographical errors. The differences remain unresolved without original programs, execution logs, or another verified source version.

5.2 All prespecified period windows

The following table reports the AIPW contrast for every declared stratum and period window. It complements the main paper's method comparison. The bootstrap applies to the primary window; alternative windows here are point comparisons. Probability contrasts are multiplied by 100 to express percentage points. These point contrasts do not become policy effects solely because several windows have the same sign; older women's pre-period trend remains a particularly relevant counterexample to that argument.

Table 2: AIPW standardized group-disparity changes across all prespecified outcome-year windows.

Stratum	Pre / post years	Weeks	Any work (pp)
Men 21–39	1987–89 / 1993–96	-2.169	-6.234
Men 21–39	1987–89 / 1995–96	-2.597	-6.950
Men 21–39	1987–91 / 1993–96	-1.864	-4.914
Men 40–58	1987–89 / 1993–96	-1.317	-2.485
Men 40–58	1987–89 / 1995–96	-0.855	-2.120
Men 40–58	1987–91 / 1993–96	-1.333	-2.538
Women 21–39	1987–89 / 1993–96	-3.553	-8.064
Women 21–39	1987–89 / 1995–96	-3.229	-7.402
Women 21–39	1987–91 / 1993–96	-3.447	-8.061
Women 40–58	1987–89 / 1993–96	-1.920	-3.472
Women 40–58	1987–89 / 1995–96	-3.166	-5.649
Women 40–58	1987–91 / 1993–96	-1.285	-2.189

5.3 Observed support in the primary window

Table 3: Primary-window support diagnostics across the four cells within each stratum.

Stratum	Min. q	Flagged (%)	Min. ESS	Ratio range	Max. w (%)
Men 21–39	0.009	0.198	2119	0.47–3.12	0.198
Men 40–58	0.025	0.000	3036	0.59–3.11	0.123
Women 21–39	0.008	0.861	1933	0.46–2.97	0.209
Women 40–58	0.020	0.000	3079	0.56–3.37	0.128

"Flagged" is the largest, across source cells, target-weight share for which that cell's joint probability is below the initial 0.01 guard; it is not a count of excluded observations. The density-ratio range removes the marginal cell-prevalence factor. The full population is retained under the documented amendment, with no trimming or probability clipping. The amendment is disclosed because replacing an initial rule after seeing the source is a design decision, even when it is motivated by support diagnostics rather than effect estimates.

The saved primary-window bootstrap summaries contain 8 stratum/outcome AIPW analyses, with 199 successful draws each and all requested draws successful. Their intervals remain conditional sampling summaries under the algorithm above, not CPS design-based or national-policy uncertainty.

6 Source verification and reproducibility

6.1 Mapping verification matrix

The matrix distinguishes verified source/publication evidence from remaining limits. The source audit is hash-bound in `data/source-verification.json` and described in `docs/source-audit.md`. Numerical comparison is a separate verification layer; neither audit supplies the missing original programs.

Table 4: Source/publication verification and its remaining limits.

Source item	Current implemented rule	Evidence and remaining limit
File and format	Supplied SAS file, loaded with Haven	Archive inventory and hashes checked; no original programs or complete-Dataverse claim
Year	Survey year 1988–1997; outcome year one less	Source labels and proof pp. 11–12 agree
Population	Survey-age 21–58; strata at 40	Source ranges and all 48 selected population/estimation counts agree
Disability	Binary `DISABL1`	Equals `DIS_HP==1` in the supplied records; interview timing and legal coverage remain distinct
Weights	Positive `FNLWGT2` / 100	Counts and numerical agreement support mapping; original reweighting code and reasons for each missing/zero weight unavailable
Employment	`WKSWORK` 0–52; zeros retained	Source label specifies last year; no missing weeks in the supplied file
Race	Source codes 3–5 pooled	Matches historical three-category specification; coding does not establish group comparability over time
Schooling and region	Three schooling groups, nine divisions	Flags verified complete/exclusive; categories reconcile the historical 162-cell definition
Earnings	Positive wages/weeks; explicit legacy trimming	All four wage-model counts agree; original CPI and selection programming cannot be inspected
Benchmarks	Independently transcribed proof targets	272 executed comparisons; discrepancies retained; final publisher printing not independently checked

Faithful replication additionally requires matching sample counts, weighted means, omitted categories, formulas, and variance calculations for explicitly selected original specifications (Acemoglu and Angrist 2001). Modernized modeling choices must be reported as extensions. A match in coefficient signs is not a replication criterion.

6.2 Commands and artifact boundaries

Run commands from the repository root after environment activation. The first block works without empirical input and validates software/report behavior:

```
source /workspace/stat256-environment/activate.sh
cd /workspace/Stat-256-Final-Project
make test
quarto render --to html
quarto render --to pdf
```

To retrieve the same supplied GitHub release archive:

```
make fetch
```

The source audit documents the available paper/data evidence and absent programs. The following interface runs the empirical pipeline on the verified SAS input:

```
Rscript scripts/run_analysis.R /absolute/path/to/marcps_w.sas7bdat \
  --bootstrap 199 --seed 256
quarto render --to html
quarto render --to pdf
```

The downloader records source metadata and checksums; originals remain under `data/raw/`. Derived observations and analysis artifacts are generated in `data/derived/` and `results/`. The paper and this appendix render to `_output/report/` as separate HTML and PDF documents. The software test fixtures live in temporary directories and must never populate research results. Retain run metadata, session information, source hashes, version-control state, and bootstrap success/failure records with any released empirical output.

`make analyze` runs the point estimates and diagnostics; `make analyze-full` adds the requested bootstrap. If a core or extension check fails, diagnose it and disclose the failed specification rather than substituting another window, trimming rule, or synthetic input. Publication requires source verification, a completed replication comparison that retains any discrepancies, executed empirical outputs, examined diagnostics, and interpretation written from those outputs.

References

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